ARITHMETIC OPERATIONS USING 8086

Aim:-

To write and execute Assembly language programs to perform arithmetic operations for the 8086 microprocessor.

Apparatus Required:-

Personal computer with MASM software.

Program:

```
ORG 1000H
MOV CL, 00H
MOV AX, 1234H
MOV BX, 1234H
ADD AX, BX
JNC L1
INC CL
L1: MOV SI, 1200H
MOV [SI], AX
MOV [SI+2], CL
MOV AH, 4CH
INT 21H
CODE ENDS
ENDS
```

Output:

Memory location	Value stored
2004H	0005H [Quotient]
2006H	0000H [Remainder]

4. DIVISION

```
ORG 1000H
MOV SI, 2000H
MOV DX, 0000H
MOV AX, [SI]
MOV BX, [SI+02H]
DIV BX
MOV [SI+04H], AX
MOV [SI+06H] AX
MOV AH, 4CH
INT 21H
CODE ENDS
END.
```

Result:-

Thus, the program to perform arithmetic operations using both direct and indirect methods are successfully executed using MASM.

Output:

Memory Location	Value stored
2004H	4444H
2006H	00H

Output:

Memory location	value store
2004 H	2468H
2006 H	0000H

2. SUBTRACTION

```
ORG 1000H
MOV SI, 2000H
MOV CL, 00H
MOV AX, [SI]
MOV BX, [SI+02H]
SUB AX, BX
JNC L1
INC CL
L1:
MOV [SI+04H], AX
MOV [SI+06H], AL
MOV AH, 4CH
INT 21H
CODE ENDS
END
```

3. MULTIPLICATION

```
ORG 1000H
MOV SI, 2000H
MOV DX, 0000H
MOV AX, [SI]
MOV BX, [SI+02H]
MUL BX
MOV [SI+04H], AX
MOV [SI+06H], AX
MOV AH, 4CH
INT 21H
CODE ENDS
END.
```

OUTPUT:

Memory location	Data stored
1200H	68H
1201H	24H
1202H	D0H